

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1712

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I Devadharshini M certify that this submission is my original work and that I ha adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Devadharshini M

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments

- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. AI-Powered Drone Delivery (Greedy Search & A*)

i. Greedy Best-First Search

Greedy Best-First Search selects the path with the smallest heuristic value (straight-line distance to destination).

Strengths

- Fast decision making.
- Low memory usage.
- Suitable for urgent situations.

Weaknesses

- Ignores actual travel cost.
- May choose risky or blocked routes.
- Does not guarantee the optimal path.

ii. A* Search

A* uses:

$$f(n) = g(n) + h(n)$$

- $g(n)$ = actual cost travelled.
- $h(n)$ = estimated cost to goal.

It balances both actual cost and heuristic estimate, helping the drone find safer and shorter routes.

iii. Impact of Heuristic Accuracy

- Accurate heuristic → Faster and better solutions.
- Overestimated heuristic → May miss optimal paths.
- Poor heuristic → More nodes explored and slower search.

iv. Effect of Dynamic Changes

- Greedy Search may repeatedly choose blocked paths.
- A* can re-evaluate costs and find alternate routes.
- Frequent updates increase computation time.

v. Recommended Algorithm

A* is the best choice because:

- Produces near-optimal routes.
- Considers safety and travel cost.
- Adapts better to changing environments.
- Provides reliable decisions for emergency medicine delivery.

2. Smart City Traffic Optimization

Optimization Formulation

Objective: Minimize

- Vehicle waiting time
- Fuel consumption
- Traffic congestion

Why Traditional Search is Inefficient

- Hundreds of intersections create a huge search space.
- Traffic changes continuously.
- Exhaustive search is computationally expensive.

i. Hill Climbing

Hill Climbing continuously modifies signal timings to improve traffic flow.

Limitations

- Can get trapped in local maxima.
- Cannot guarantee global optimum.
- Sensitive to starting conditions.

ii. Local Maxima, Plateaus and Ridges

Local Maxima

- A solution better than nearby solutions but not globally best.

Plateau

- Large region with similar traffic performance.
- No clear direction for improvement.

Ridge

- Improvement requires multiple coordinated changes.
- Single-step changes may fail.

iii. Simulated Annealing / Genetic Algorithm

Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.

Genetic Algorithm

- Maintains multiple solutions.
- Uses selection, crossover, and mutation.
- Explores large search spaces efficiently.

iv. Recommended Approach

Genetic Algorithm

Reasons

- Highly scalable.
- Adapts to changing traffic conditions.
- Finds high-quality solutions in large search spaces.

3. Mars Rover Exploration

i. Why Online Search?

The rover does not have a complete map.

It must:

- Sense environment.
- Move.
- Learn new information.

Therefore, online search is required.

ii. Challenges

Partial Observability

- Limited sensor range.
- Unknown obstacles.

Dynamic Environment

- Terrain conditions may change.
- New hazards may appear.

iii. Internal Model Update

The rover:

1. Collects sensor data.
2. Maps explored regions.
3. Updates obstacle locations.
4. Stores safe routes.
5. Uses new information for future decisions.

iv. Online vs Uninformed vs Informed Search

Feature	Online Search	Uninformed Search	Informed Search
Complete Map	Not Required	Required	Required
Adaptability	High	Low	Medium
Real-Time	Yes	No	Limited
Learning	Yes	No	No

v. Recommended Approach

Online Search Agent

Reasons

- Handles uncertainty.
- Learns continuously.
- Adapts to unknown environments.
- Ensures safer navigation.

4. Exam Timetable Generation (CSP)

CSP Formulation

Variables

- Course exams

Domains

- Available time slots
- Available exam halls

Constraints

1. No student overlap.
2. Hall capacity limits.
3. Faculty availability.
4. Subject precedence rules.
5. Special accommodation requirements.

ii. Backtracking Search

1. Assign a time slot.
2. Check constraints.
3. If violation occurs, backtrack.
4. Try another assignment.
5. Continue until solution is found.

iii. Constraint Propagation (Forward Checking)

- Removes invalid choices early.
- Reduces search space.
- Detects conflicts quickly.
- Improves efficiency significantly.

iv. Real-World Challenges

- Large number of courses.
- Thousands of students.
- Limited halls.
- Dynamic faculty schedules.

Best Strategy

Backtracking + Forward Checking

Reasons

- Reduces computation.
- Produces valid schedules efficiently.
- Scales well for large universities.

5. Strategic Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax Algorithm

Minimax assumes:

- AI tries to maximize score.
- Opponent tries to minimize score.

The AI selects moves that provide the best worst-case outcome.

ii. Computational Challenges

- Huge number of possible moves.
- Large game tree depth.
- High memory consumption.
- Slow decision making.

iii. Alpha-Beta Pruning

Alpha: Best value for MAX player.

Beta: Best value for MIN player.

If:

$\text{Alpha} \geq \text{Beta}$

the remaining branch is ignored.

Benefits

- Reduces nodes explored.
- Produces same result as Minimax.
- Faster computation.

iv. Evaluation Function

Factors:

- Resource count.

- Army strength.
- Territory control.
- Defensive positions.
- Unit health.
- Strategic advantage.

Example

Evaluation Score =
 $(3 \times \text{Resources}) +$
 $(5 \times \text{Army Strength}) +$
 $(4 \times \text{Territory Control})$

v. Recommended Strategy

Minimax + Alpha-Beta Pruning + Heuristic Evaluation

Reasons

- Maintains strong decision quality.
- Reduces computation.
- Meets real-time constraints.
- Provides competitive gameplay.